

Weekly Self-Study Tasks to Complete

AI Code Generation with GitHub Copilot and Liberty GPT for Leaders

This document covers all of Week 5 of Phase 3 in the 4-hour Leaders model: 2 hours of instructor-led session, 1 hour of self-study video, and 1 hour of hands-on lab. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section points to the Evergreen AI-assisted feature lab, where you will use Copilot's inline completions and its Ask, Edit, Plan, and Agent chat modes to build features, catch AI mistakes, and verify with tests. Additional Content at the end is optional and sits outside the 4-hour budget.

Core Resources (~60 min)

Watch the specific resources called out below — not an entire course. Keep a critical eye: for each AI suggestion the videos show, ask whether you would trust it.

Topic	Material Description	Est. Time	Link / Reference
How AI Coding Works	AI Pair Programming with GitHub Copilot (Ronnie Sheer, LinkedIn Learning, released Apr 3, 2025; 1h 32m course). Watch “Getting Familiar with Copilot” (~25 min). What AI assistants do, how context shapes suggestions, and their limits.	~25 min	linkedin.com/learning — AI Pair Programming with GitHub Copilot
Copilot in Practice	Quickstart for GitHub Copilot (official GitHub docs, VS Code tab) — sign-in, inline completions, and Copilot Chat. Read and skim the workflow.	~18 min	docs.github.com — Copilot in VS Code
Prompt Engineering	Prompt engineering for GitHub Copilot (official GitHub docs) — a	~17 min	docs.github.com — Prompt engineering

Topic	Material Description	Est. Time	Link / Reference
	focused guide to writing clear prompts (context, task, constraints).		

Practice Activities (1 hour)

Complete this week's hands-on lab: "Evergreen Insurance — Building Features with AI Assistance, Responsibly." Working with GitHub Copilot in VS Code, you will use inline completions and all four chat modes (Ask, Edit, Plan, and Agent) to generate a feature from a prompt, catch an AI-introduced bug, spot a hallucinated API, review and delegate work to an agent, and verify everything with tests. The whole exercise takes about one hour.

In the lab you will:

- Open the Labfiles/evergreen-quote-copilot-assist folder in VS Code and run `node --test` in the integrated terminal to see the starting state.
- Generate `applyMultiPolicyDiscount` from the specification comment using inline completions.
- Use Ask mode to diagnose the plausible-but-wrong bug in `formatQuoteSummary`, then fix it.
- Spot the hallucinated API in `parseAmount`, have Copilot debunk it, and replace it with real code.
- Rewrite a weak prompt into a strong one using Context, Task, Constraints, and Format, and run both in Ask mode to compare the results.
- Explain the `riskTier` function with Ask mode, then refactor it safely with Edit mode, using the tests as a safety net.
- Review a Plan-mode implementation plan for `quoteBundleTotal`, then delegate the implementation to Agent mode and verify its diff and the tests yourself.
- Re-run `node --test` in the integrated terminal until everything is green, and reflect on responsible enterprise use.

The lab instructions (with "Why this matters" callouts throughout) are in the lab environment. Budget one hour and work at your own pace.

Additional Content (Optional)

The following resources are optional and recommended for students who want to go deeper beyond the core curriculum. They are outside the 4-hour weekly budget.

Focus this week on how to direct an AI tool well and how to verify its output.

Topic	Material Description	Link / Reference
Copilot	GitHub Copilot documentation — the authoritative guide to features, configuration, and best practices (Reference).	docs.github.com/copilot
Responsible AI	Microsoft Responsible AI — principles and controls for safe, accountable enterprise AI use (Reference).	microsoft.com/ai/responsible-ai
Prompting	Prompt Engineering Guide — a broad, practical catalog of prompt patterns and strategies (Reference).	promptingguide.ai
Hands-On	GitHub Skills — free, guided interactive courses, including using GitHub Copilot (Interactive).	skills.github.com
Testing with AI	Writing tests with GitHub Copilot — official guidance on using AI for test scaffolding, and why you must expand and verify coverage (Tutorial).	docs.github.com — Copilot tests